Identification of Landmarks for Streamlining Existing CD45/SSC Lineage Manual Gating Operations

Christopher Snyder MD, PhD* and Cara Shirai PhD**
Department of Pathology & Immunology
Washington University in St. Louis, St. Louis, MO

* csnyder0@uw.edu
Current Hematopathology Fellow at University of Washington; work performed at Washington University in St. Louis.

** Medical Director of Flow Cytometry at WashU

SETUP

ClearLab10C [1], New Patient Screen, BJH

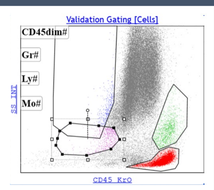
	Blue laser (488 nm)					Red laser (638 nm)			Violet laser (405 nm)	
	FITC	PE	ECD	PE-Cy5.5	PE-Cy7	APC	APC-A700	APC-A750	PacB	KrO
B Cell Tube	Kappa	Lambda	CD10	CD5	CD200	CD34	CD38	CD20	CD19	CD45
T Cell Tube	TCRγδ	CD4	CD2	CD56	CD5	CD34	CD7	CD8	CD3	CD45
M1 Cell Tube	CD16	CD7	CD10	CD13	CD64	CD34	CD14	HLA-DR	CD11b	CD45
M2 Cell Tube	CD15	CD123	CD117	CD13	CD33	CD34	CD38	HLA-DR	CD19	CD45

New Patient Screen Assay Beckman Coulter 4 x 10-Color Tubes

CHALLENGE

Initial Lineage Gate Assignment (CD45 x SSC)

Rate-Limiting
Single Point of Failure



SIGNIFICANCE

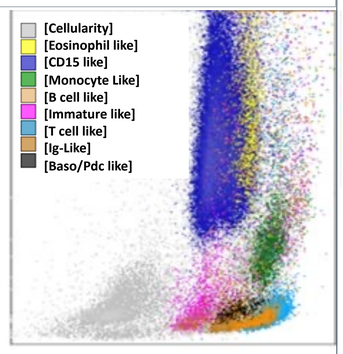
Index Gates are Automated Visual Cues

Key Properties
-Aberrancy tolerant

- Cross-tube correlations (CD45xSSC alignment)

- Decreased inter-user gate placement variability

- 2D "Snapshot at a glance"
- Intuitive preset models
  - "Book" patterns



4-tube Index Gate Composite

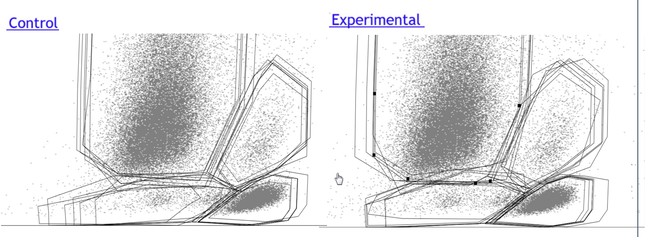
STUDY DESIGN

Participants: Six technologists
Design: 100 Normal Retrospective + Abnormal Case Series
Protocol intervention: ± Visual: Index Gate Composite
Retrospective Design:
Within-subject crossover control,
2 week washout two-period internal
Outcomes measured:
Correlation with values from final reports
Percent gating error relative to reference values
Result:
1. Noninferior on normal clinical flow data
2 Improved gate placement consistency (subjective assessment)

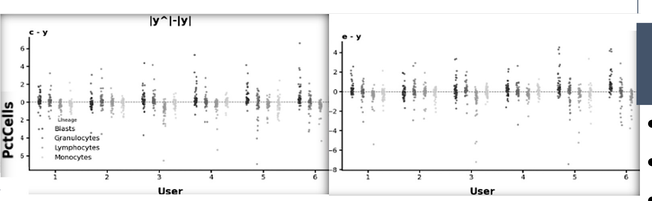
RESULTS

Normal Data, Retrospective Analysis

Control Experimental

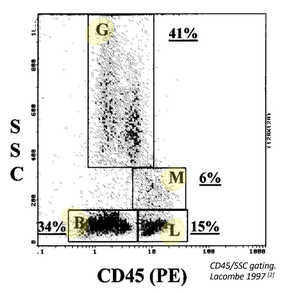


- Unaided CD45dim gates are more dispersed (control)



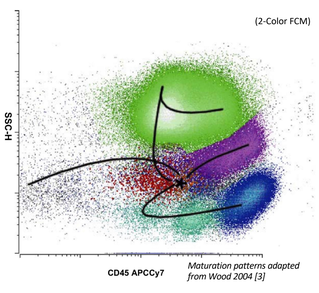
- Gate percent cellularity vs reported was not significantly different across methods

PERSPECTIVES AND PREVIOUS WORK



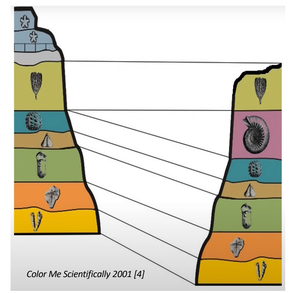
CD45/SSC gating. Lacombe 1997 [2]

CD45 x SSC was introduced as superior to SSC x FSC for lineage separation. Demonstrating the power of expression data.



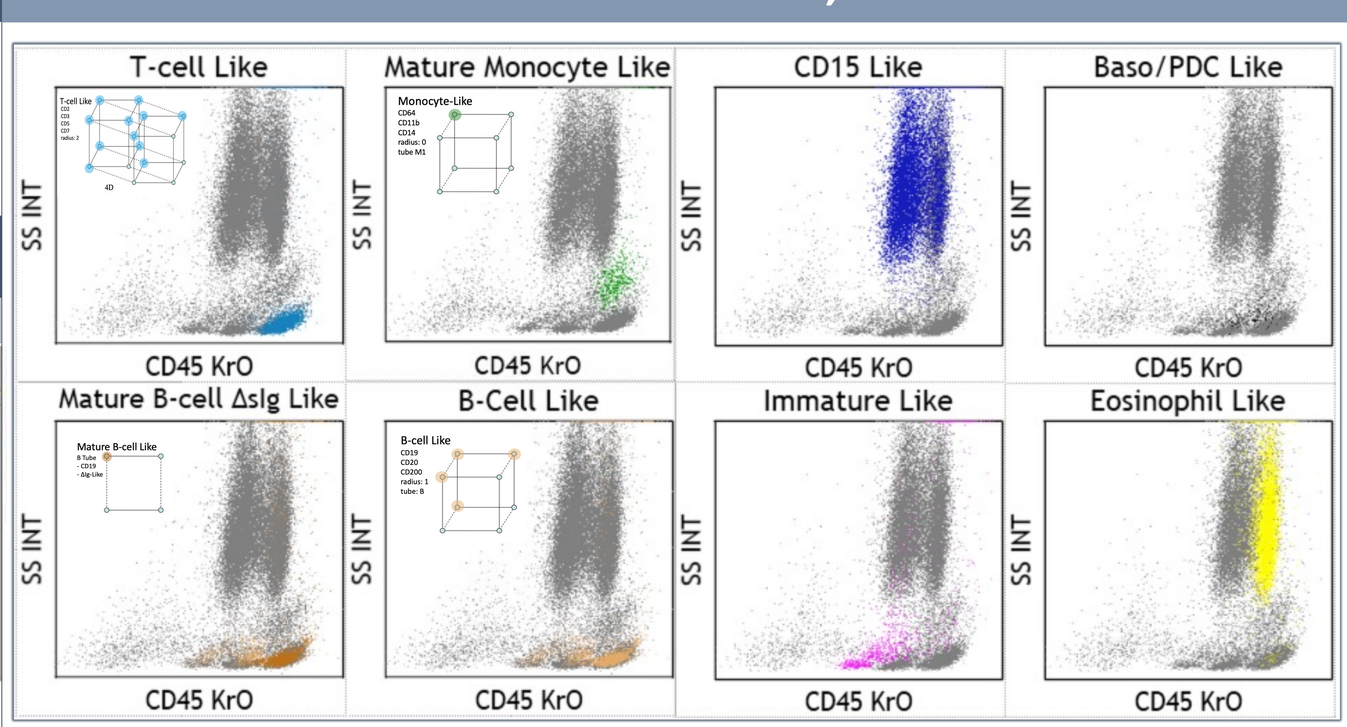
Maturation patterns adapted from Wood 2004 [3]

Maturation curves: An *atlas* for phenotype space. Conversely, phenotypes are often *geographically* specific.



An *Index Fossil* tells you *where you are* by *how it is.* An *Index Gate* is a phenotype defined landmark..

INDEX GATE COMPONENTS, A CASE STUDY

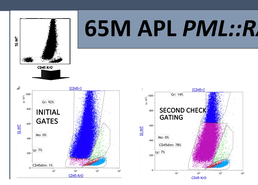


- A 3-year-old boy s/p allergy testing demonstrates cell subsets nicely

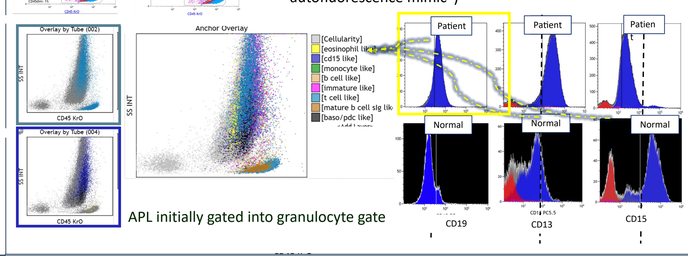
RESULTS

Abnormal Data, Case Series

65M APL *PML::RARA*+ APL in Gr Gate BM **APL - like**

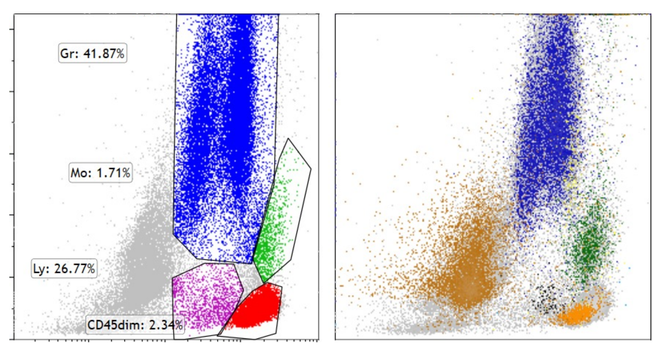


- Notice how few Grans are CD15-like! (APL negative for)
- Instead many are "T cell-like" (teal, due to **CD2** and **CD7**)
- "Eos-like" unusual placement (yellow, due to CD15(-) "autofluorescence mimic")

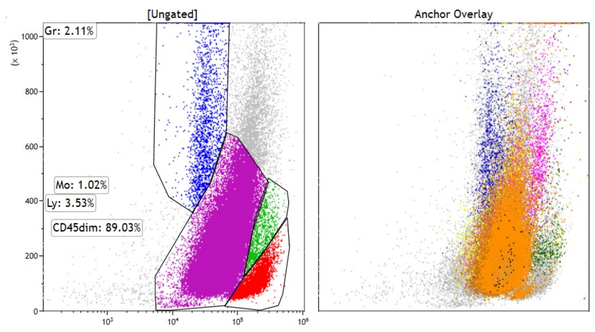


APL initially gated into granulocyte gate

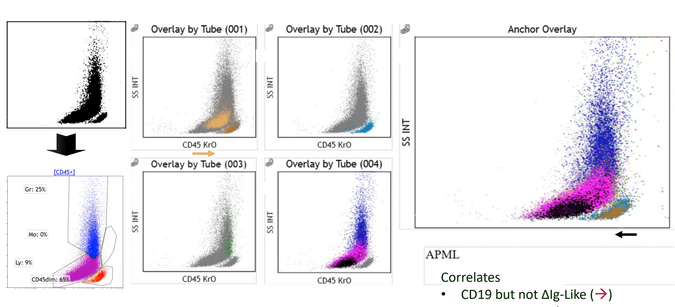
61M new IgG κ Multiple Myeloma



11M Burkitt's Lymphoma **Δ I, like**



76M new AML, t(8;21) PBL **AML with RUNX1T1::RUNX1**



APML

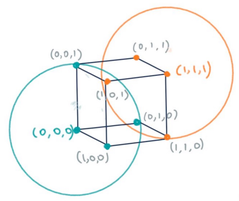
Correlates
- CD19 but not ΔIg-Like (→)
- CD34, HLA-DR (←) Immature-like

CONCLUSIONS

- Index Gates are an accessible way to profile multiparametric data.
- Aberrancy tolerance is also robustness to imprecise gating.
- Collective knowledge "Cannon phenotypes" can be a basis for interpretable complex models.

Aberrancy Radius Gating

- A mathematical means for building "slack" into AND gating
- Motivated by error correcting codes

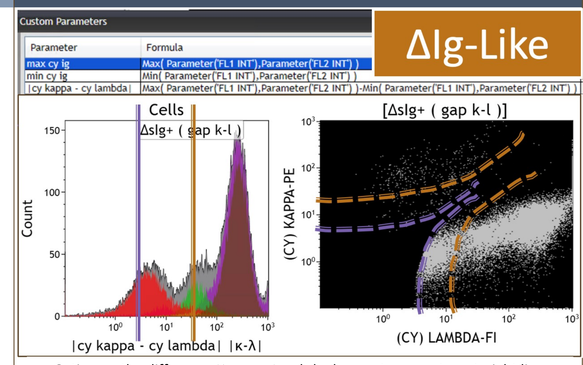



```
Immature-Like {HLA-DR, CD34, CD38, CD117}C3

CD117: Parameter( 'FL3 INT' )
CD34: Parameter( 'FL6 INT' )
CD38: Parameter( 'FL7 INT' )
HLA-DR: Parameter( 'FL8 INT' )
|
Max(
    Max(
        Min(
            Min(Parameter('FL8 INT'), Parameter('FL6 INT')),
            Parameter('FL7 INT')
        ),
        Min(
            Min(Parameter('FL8 INT'), Parameter('FL6 INT')),
            Parameter('FL3 INT')
        )
    ),
    Max(
        Min(
            Min(Parameter('FL8 INT'), Parameter('FL7 INT')),
            Parameter('FL3 INT')
        ),
        Min(
            Min(Parameter('FL6 INT'), Parameter('FL7 INT')),
            Parameter('FL3 INT')
        )
    )
)
```

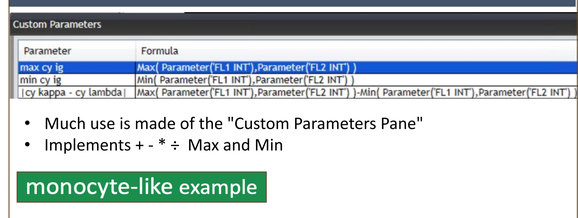
Light Chain Gap

ΔIg-Like



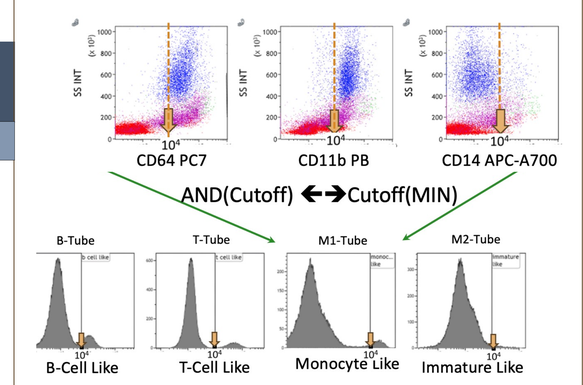
- Gating on the difference Kappa – Lambda does not map onto a straight line

Kaluza Configuration



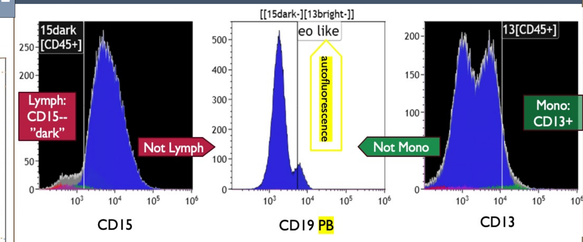
- Much use is made of the "Custom Parameters Pane"
- Implements + - * ÷ Max and Min

monocyte-like example



- Numerical multi-channel comparisons made possible by common calibration
- *Simply, "Max" and "Min" are continuous-valued versions of "OR" and "AND"*
- *A single free parameter can thus be used to "cutoff" multi parameter models*

"Not Unlike" Eosinophils



- Eosinophil model relies on ruling out other populations
- Followed by leveraging granule autofloruescence properties

ACKNOWLEDGEMENTS AND REFERENCES

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[3] Wood, Brent. "Multicolor immunophenotyping: human immune system hematopoiesis." *Methods in cell biology*. Vol. 75. Academic Press, 2004. 559-576.

[4] Color Me Scientifically. (2020, July 30). Index fossils [Video]. YouTube. https://www.youtube.com/watch?v=OD2coZmSZbI